

Mehrad Ansari

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AI research scientist building autonomous agents and tools for scientific discovery, experience in LLM post-training (SFT/RL), production agent and multi-modal RAG infrastructure, and small molecule design across drug and materials discovery.

EDUCATION

Ph.D., Chemical Engineering University of Rochester	2019 - 2023
<i>Thesis: Applications of Physics-informed Machine Learning in Chemical Engineering (Advisor: Andrew D. White)</i>	
M.S., Chemical Engineering University of Rochester	2019 - 2021
M.S., Environmental Engineering Missouri University of Science & Technology	2016 - 2018
<i>Thesis: Numerical Modeling of Capillary-driven Flow in Open Microchannels: An Implication of Optimized Wicking Fabric Design</i>	
B.S., Chemical Engineering University of Tehran	2010 - 2015
<i>Thesis: Experimental Setup and Optimization for Electro-catalytical Generation of Hydroxyl Radicals in Wastewater Treatment</i>	

EXPERIENCE

Senior Scientist

June 2025 – Present

Lila Sciences, Boston, MA

- Designed scientific AI agents with ReAct reasoning, long-term memory, and tool use for long-horizon inverse small molecule design tasks, coupling LLM reasoning with physics-informed surrogate models; built stateful code-execution sandboxes for autonomous experimentation.
- Fine-tuned open-weight LLMs (Mistral-7B) with supervised fine-tuning and GRPO (*reinforcement learning from verifiable rewards*) for constrained SMILES molecular generation on EC2 GPU instances.
- Built a scalable scientific-literature multi-modal RAG service exposed via an MCP endpoint on Kubernetes/AWS; designed evaluation benchmarks (LitQA2) and optimized hybrid BM25+dense retrieval with reranking, benchmarked against PaperQA2.
- Built a therapeutic bioactivity-prediction system for EGFR inhibitor potency (IC50) using ChEMBL data.
- Led and mentored cross-functional teams of PhD-level researchers and engineers across materials and drug discovery campaigns

Staff Scientist

Jan 2024 – June 2025

Acceleration Consortium Research Fellow

June – Dec 2023

Acceleration Consortium, University of Toronto, Toronto, ON

- Lead developer of **dZiner**, an LLM multi-agent framework for interpretable, literature-grounded inverse molecular design. Designed *de novo* small-molecule inhibitors of the WDR5 oncology target; top candidates were synthesized (WuXi AppTec) and experimentally validated via fluorescence-polarization displacement, affinity-selection mass spectrometry, and differential scanning fluorimetry ([manuscript under review](#), *Nature Machine Intelligence*).
- Applied representation learning, Bayesian optimization, and AI agents to accelerate self-driving chemistry-lab campaigns and hit-to-lead optimization with built-in domain knowledge.
- Developed chemistry-informed AI agents (Slack API) and agent-based LLM plugins to accelerate design of materials for CO₂ reduction.
- Built **Eunomia**, an agent-based system for large-scale extraction of materials datasets from scientific literature (*Digital Discovery*).
- Mentored graduate and undergraduate students and co-organized global hackathons on AI for chemistry and materials.

Research Assistant

2019 – 2023

University of Rochester, Rochester, NY

- [HuggingFace literature-QA app](#) that answers questions from scientific papers using LLMs, an early retrieval-augmented scientific assistant.
- Developed [Peptide.bio](#), bidirectional-LSTM sequence-to-function models predicting therapeutic peptide properties (hemolysis, solubility, non-fouling, SHP-2 target binding) directly from amino-acid sequence; introduced a positive-unlabeled learning method to train on positives-only biological screening data, deployed serverless in-browser as edge-computing ML
- Built disease-modeling ([py0](#)) and active-learning CFD / symbolic-regression tools ([AL-CFD](#)) for scientific simulation.

Energy & Materials ML Research Engineer / Intern

May 2022 – Mar 2023

Toyota Research Institute, Los Altos, CA

- Developed deep-learning architectures to predict degradation of used Li-ion batteries with unknown cycling histories (**US Patent pending**).

Manufacturing Process Modeling Intern

May – Dec 2017

The Goodyear Tire & Rubber Company, Akron, OH

- Modeled and optimized the tire-vulcanization process (phase-change heat transfer) in ANSYS; automated simulation workflows (OPTIMUS, MATLAB) and validated against plant data using adaptive mesh refinement on HPC.

Missouri University of Science & Technology, Rolla, MO

- Numerical modeling of multiphase flow in open microfluidics (ANSYS, STAR-CCM+); developed adaptive mesh-refinement algorithms.

SELECTED PUBLICATIONS & PATENTS (see full list on [Google Scholar](#))

1. [dZiner: Rational Inverse Design of Materials with AI Agents](#) **Project Lead** Spotlight, AI4Mat-NeurIPS (2024), *Nature Machine Intelligence* (under review 2026)
M. Ansari, J. Watchorn, C. E. Brown, J. S. Brown
2. [Agent-based Learning of Materials Datasets from Scientific Literature \(Eunomia\)](#) *Digital Discovery* (2024)
M. Ansari, S. M. Moosavi
3. [Learning Peptide Properties with Positive Examples Only](#) *Digital Discovery* (2024)
M. Ansari, A. D. White
4. [32 Examples of LLM Applications in Materials Science and Chemistry](#) **Project co-organizer & team lead** *MLST* (2025)
Y. Zimmermann, A. Bazgir, A. Al-Feghali, M. Ansari, et al.
5. [Kernel-Learning-Assisted Synthesis Condition Exploration for Ternary Spinel](#) **Project co-lead** *Nature Communications Materials* (2025)
Y. Liu, M. Ansari, R. Black, J. Hatrick-Simpers
6. [History-agnostic Battery Degradation Inference](#) *Journal of Energy Storage* (2024) · US Patent Pending
M. Ansari, S. Torrisi, A. Trewartha, S. Sun
7. [Serverless Prediction of Peptide Properties with Recurrent Neural Networks](#) *Journal of Chemical Information and Modeling* (2023)
M. Ansari, A. D. White
8. [Assessment of Chemistry Knowledge in Large Language Models that Generate Code](#) *Digital Discovery* (2023)
A. D. White, G. M. Hocky, H. A. Gandhi, M. Ansari, S. Cox, G. P. Wellawatte, S. Sasmal, Z. Yang, K. Liu, Y. Singh, W. J. Peña Ccoa
9. [Bayesian Optimization Hackathon for Chemistry and Materials](#) **Project co-lead** *chemRxiv* (2025)
S. Baird, M. Ansari, Z. Afzal, Q. Ai, et al.
10. [Iterative Symbolic Regression for Learning Transport Equations](#) *AIChE Journal* (Invited special Issue on AI 2022)
M. Ansari, H. A. Gandhi, D. G. Foster, A. D. White
11. [Inferring Spatial Source of Disease Outbreaks using Maximum Entropy](#) *Physical Review E* (2022)
M. Ansari, D. Soriano-Paños, G. Ghoshal, A. D. White
12. [Simulation-based Inference with Approximately Correct Parameters via Maximum Entropy](#) *NeurIPS* (2020), *MLST* (2022)
R. Barrett, M. Ansari, G. Ghoshal, A. D. White

SELECTED TALKS

- Teaching Computers How to Do Science (**Invited**) — IBM Research, Tokyo, Japan Feb 2025
- Rational Inverse Design of Materials with Chemist AI Agents — Spotlight, AI4Mat-NeurIPS 2024, Vancouver, BC Dec 2024
- Materials Inverse Design with AI Agents (**Invited**) — Trillion Parameter Consortium, Barcelona, Spain June 2024
- Multi-modal AI Agents in Materials Discovery — Accelerate 24, Vancouver, BC Aug 2024
- Flexible Automation of Self-driving Labs with Built-in Domain Knowledge (**Invited**) — Toyota Research Institute, Los Altos, CA Mar 2024
- Positive Unlabeled Learning of Peptide Properties — Accelerate23, Toronto, ON Aug 2023

HONORS & AWARDS

- Acceleration Consortium Research Fellowship, University of Toronto 2023
- 1st Place — Battery Informatics & ML Kaggle Competition, Materials Research Society 2022
- 3rd Place (Radical AI) — LLM Hackathon for Materials & Chemistry, Toronto 2024
- 2nd Place — Bayesian Optimization Hackathon for Chemistry & Materials, Toronto 2024
- Kwang-Yu and Lee-Chien Wang Fellowship, University of Rochester 2021
- Earl W. Costich Graduate Fellowship, University of Rochester 2020
- Recognized Reviewer, Journal of Environmental Chemical Engineering 2016

TECHNICAL SKILLS

Scientific Software: [dZiner](#) · [Eunomia](#) · [Peptide.bio](#) · [MaxEnt](#) · [Py0](#) · [AL-CFD](#) · [HOOMD-TF](#)

LLM & agents: RAG, multi-agent frameworks, ReAct, MCP, RLHF/GRPO, SFT, eval harnesses (LitQA2)

ML / cheminfo: PyTorch, TensorFlow, JAX, scikit-learn, LiteLLM, vLLM, RDKit; Bayesian optimization, surrogate modeling

Data & DBs: ChEMBL, PubChem, PDB

Infrastructure: AWS, Kubernetes, Ray, HPC/Slurm, Lambda Labs

Languages: Python, JavaScript, HTML/CSS